DR. RVR NRI DEEMED TO BE UNIVERSITY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

PYTHON PROGRAMMING
ACTIVITY-BASED LEARNING (ABL) ASSIGNMENT

Student Name	CH. DURGA PRASAD SESHADRI
Roll Number	26000016
Branch & Section	CSE - 1
Academic Year	2026–2027

ASSIGNMENT OVERVIEW

This report contains ten Python programming tasks. Each task includes its task heading, the submitted Python source code, and one compact flowchart.

TASK INDEX

No.	Task Name
1	Attendance and Extra Classes Calculator
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Task 1: Attendance and Extra Classes Calculator

PYTHON SOURCE CODE

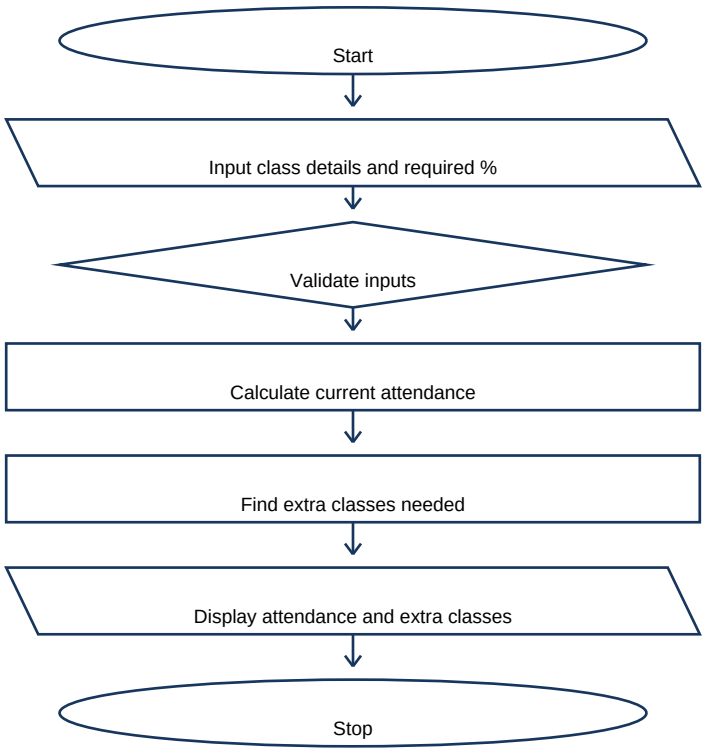
```
total_classes = int(input("Enter total classes held: "))
attended = int(input("Enter classes attended: "))
required = float(input("Enter required attendance percentage: "))

if total_classes <= 0 or attended < 0 or attended > total_classes:
    print("Invalid attendance details")
elif not 0 < required <= 100:
    print("Invalid required percentage")
else:
    percentage = attended / total_classes * 100
    print("Current attendance:", round(percentage, 2), "%")

    future = 0
    while (attended + future) / (total_classes + future) * 100 < required:
        future += 1

    print("Additional classes required:", future)
```

FLOWCHART



OUTPUT

```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

======= RESTART: C:/Users/chitt/OneDrive/Desktop/python git hub/task 1.py ======
Enter total classes held: 80
Enter classes attended: 50
Enter required attendance percentage: 30
Current attendance: 62.5 %
Additional classes required: 0
```

Task 2: Bank Account Management System

PYTHON SOURCE CODE

```
balance = 5000
transactions = []

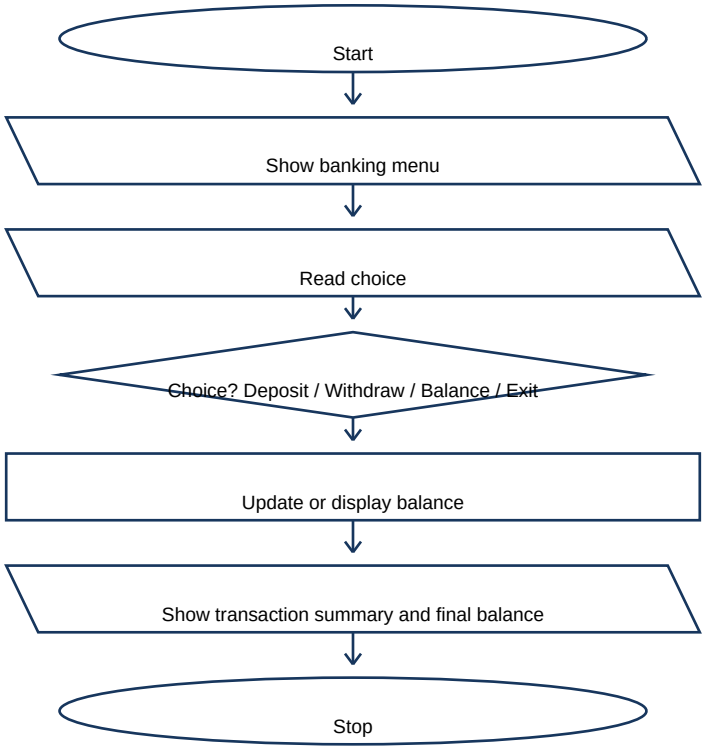
while True:
    print("\n1. Deposit")
    print("2. Withdraw")
    print("3. Check Balance")
    print("4. Exit")

    choice = input("Enter your choice: ")

    if choice == "1":
        amount = float(input("Enter deposit amount: "))

        if amount > 0:
            balance += amount
            transactions.append(("Deposit", amount))
            print("Deposit successful")
        else:
            print("Invalid amount")
```

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OUTPUT

```
1. Deposit
2. Withdraw
3. Check Balance
4. Exit
Enter your choice: 2
Enter withdrawal amount: 5000
Withdrawal successful

1. Deposit
2. Withdraw
3. Check Balance
4. Exit
Enter your choice:
```

Task 3: Exam Performance and Improvement Tracker

PYTHON SOURCE CODE

```
marks = []

for i in range(3):
    score = float(input("Enter exam " + str(i + 1) + " marks: "))

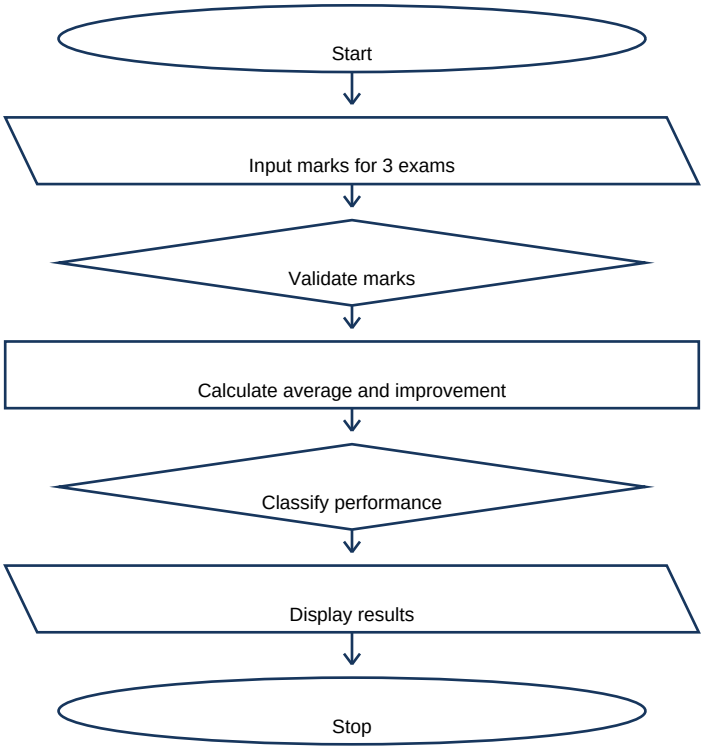
    if score < 0 or score > 100:
        print("Invalid marks")
        break

    marks.append(score)

if len(marks) == 3:
    average = sum(marks) / len(marks)

    print("Average marks:", round(average, 2))
    print("Improvement:", marks[2] - marks[0])
```

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OUTPUT

```
Enter exam 1 marks: 35
Enter exam 2 marks: 40
Enter exam 3 marks: 45
Average marks: 40.0
Improvement: 10.0
Performance: Needs Improvement
```

Task 4: Travel Cost Comparison for Vehicles

PYTHON SOURCE CODE

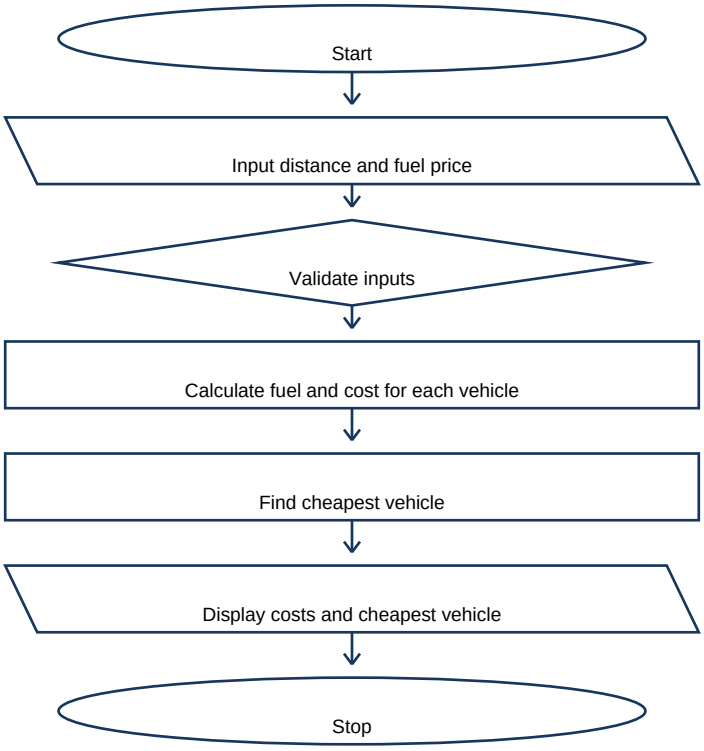
```
distance = float(input("Enter journey distance in km: "))
fuel_price = float(input("Enter fuel price per litre: "))

vehicles = [
    ("Bike", 40),
    ("Car", 15),
    ("Mileage Car", 20)
]

costs = []

if distance <= 0 or fuel_price <= 0:
    print("Enter valid distance and fuel price")
else:
    for name, mileage in vehicles:
        fuel = distance / mileage
        cost = fuel * fuel_price
```

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OUTPUT

```
Enter journey distance in km: 500
Enter fuel price per litre: 117
Bike
Fuel required: 12.5 litres
Travel cost: Rs. 1462.5
Car
Fuel required: 33.33 litres
Travel cost: Rs. 3900.0
Mileage Car
Fuel required: 25.0 litres
Travel cost: Rs. 2925.0
Cheapest vehicle: Bike
Minimum cost: Rs. 1462.5
```

Task 5: Number Pattern Printing

PYTHON SOURCE CODE

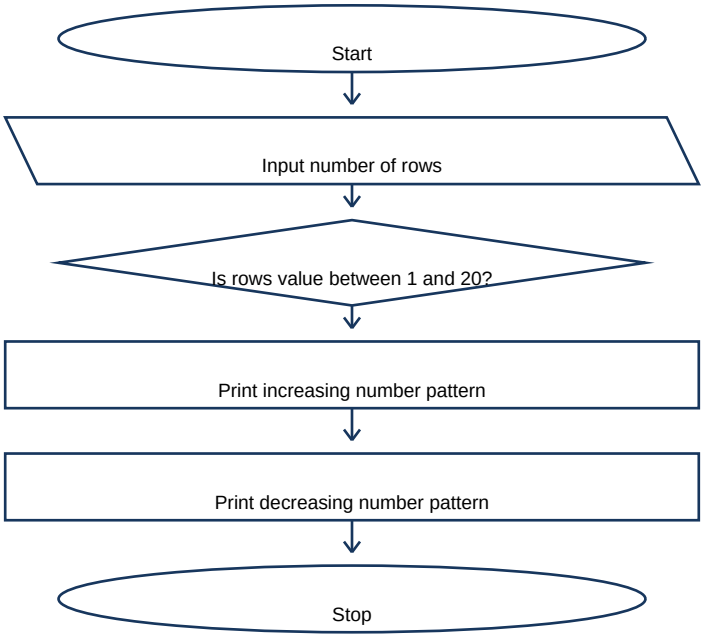
```
n = int(input("Enter number of rows: "))

if n < 1 or n > 20:
    print("Enter a number between 1 and 20")
else:
    for i in range(1, n + 1):
        for j in range(1, i + 1):
            print(j, end=" ")
        print()

    for i in range(n - 1, 0, -1):
        for j in range(1, i + 1):
            print(j, end=" ")
        print()
```

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OUTPUT

```
Enter number of rows: 4
1
1 2
1 2 3
1 2 3 4
1 2 3
1 2
1
```

Task 6: Password Strength Validator

PYTHON SOURCE CODE

```
password = input("Enter your password: ")

if len(password) < 8:
    print("Password must contain at least 8 characters")

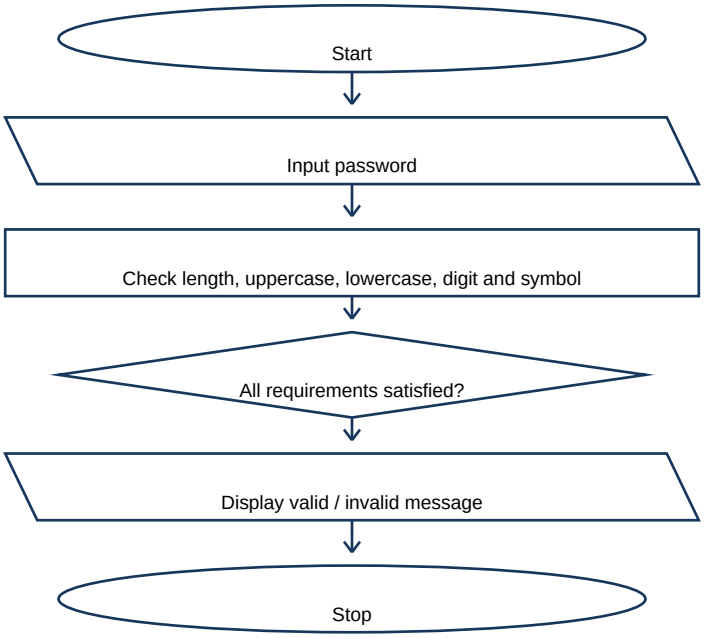
if not any(c.isupper() for c in password):
    print("Add an uppercase letter")

if not any(c.islower() for c in password):
    print("Add a lowercase letter")

if not any(c.isdigit() for c in password):
    print("Add a digit")

if not any(not c.isalnum() for c in password):
    print("Add a special character")
```

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OUTPUT

```
Enter your password: 1234567
Password must contain at least 8 characters
Add an uppercase letter
Add a lowercase letter
Add a special character
Password is invalid

======= RESTART: C:/Users/chitt/OneDrive/Desktop/python git hub/task 6.py ======
Enter your password: 03Jan2009@
Password is valid
```

Task 7: String Examination Answer Evaluation

PYTHON SOURCE CODE

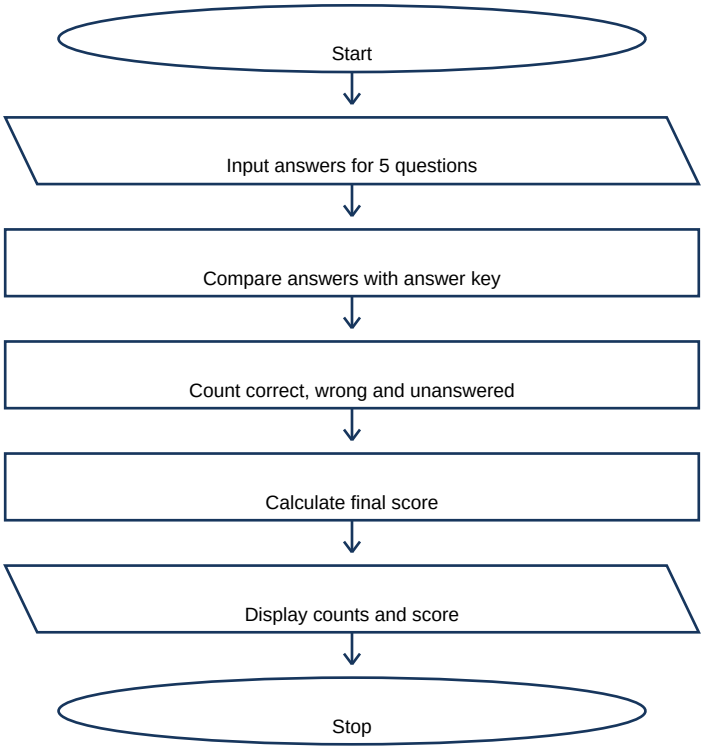
```
key = ["A", "B", "C", "D", "A"]
answers = []

for i in range(len(key)):
    answer = input("Enter answer for question " + str(i + 1) + ": ")
    answers.append(answer.strip().upper())

correct = 0
wrong = 0
unanswered = 0

for i in range(len(key)):
    if answers[i] == "":
        unanswered += 1
    elif answers[i] == key[i]:
        correct += 1
    else:
```

FLOWCHART



OUTPUT

```
Enter answer for question 1: 15
Enter answer for question 2: 3
Enter answer for question 3: 7
Enter answer for question 4: 9
Enter answer for question 5: 4
Correct answers: 0
Wrong answers: 5
Unanswered: 0
Final score: -2.5 out of 10
```

Task 8: String Compression Analyzer

PYTHON SOURCE CODE

```
text = input("Enter a string: ")

compressed = ""
i = 0

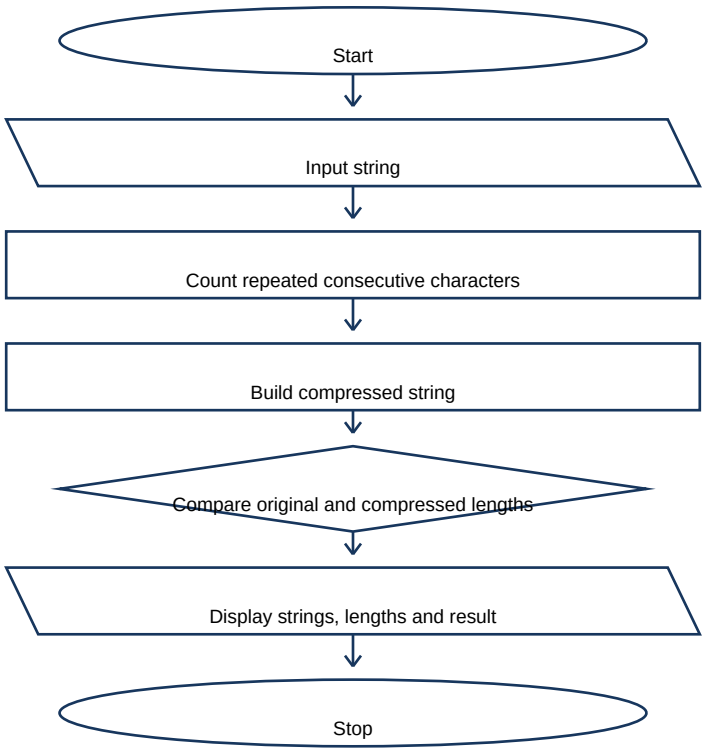
while i < len(text):
    count = 1

    while i + count < len(text) and text[i] == text[i + count]:
        count += 1

    compressed += text[i] + str(count)
    i += count

print("Original text:", text)
print("Compressed text:", compressed)
print("Original length:", len(text))
```

FLOWCHART



OUTPUT

```
Enter a string: durga prasad seshadri
Original text: durga prasad seshadri
Compressed text: d1u1r1g1a1 1p1r1a1s1a1d1 1s1e1s1h1a1d1r1i1
Original length: 21
Compressed length: 42
Compression is not beneficial
```

Task 9: Student Marks and Subject Average Analyzer

PYTHON SOURCE CODE

```
students = ["Ravi", "Sita", "Arun"]
subjects = ["Python", "Maths", "English"]

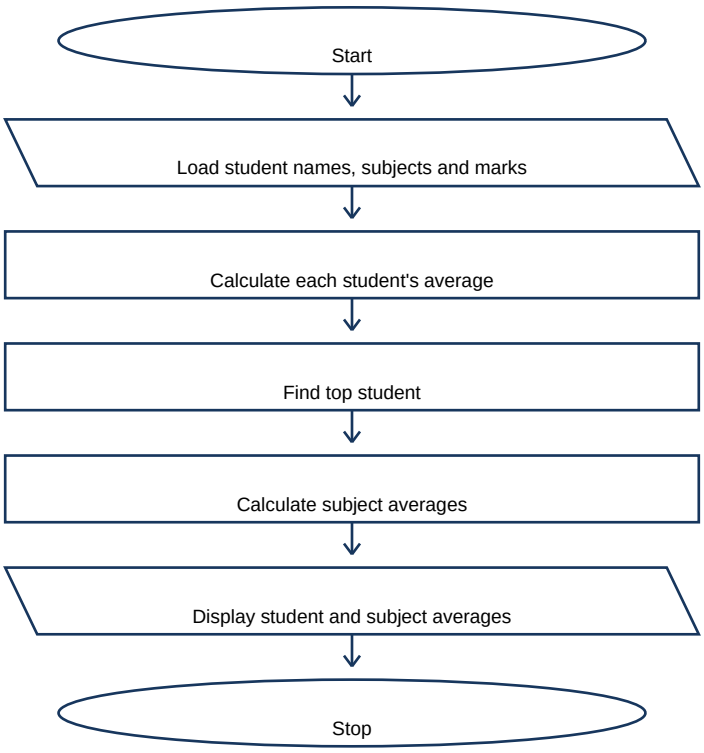
marks = [
    [80, 75, 90],
    [90, 85, 95],
    [70, 80, 75]
]

averages = []

for i in range(len(students)):
    average = sum(marks[i]) / len(marks[i])
    averages.append(average)
    print(students[i], "Average:", round(average, 2))

top_index = averages.index(max(averages))
```

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OUTPUT

```
Ravi Average: 81.67
Sita Average: 90.0
Arun Average: 75.0
Top student: Sita
Python Average: 80.0
Maths Average: 80.0
English Average: 86.67
```

Task 10: Food Wastage Tracker

PYTHON SOURCE CODE

```
days = ["Monday", "Tuesday", "Wednesday", "Thursday", "Friday"]
prepared = [100, 120, 110, 90, 130]
served = [90, 100, 105, 70, 115]

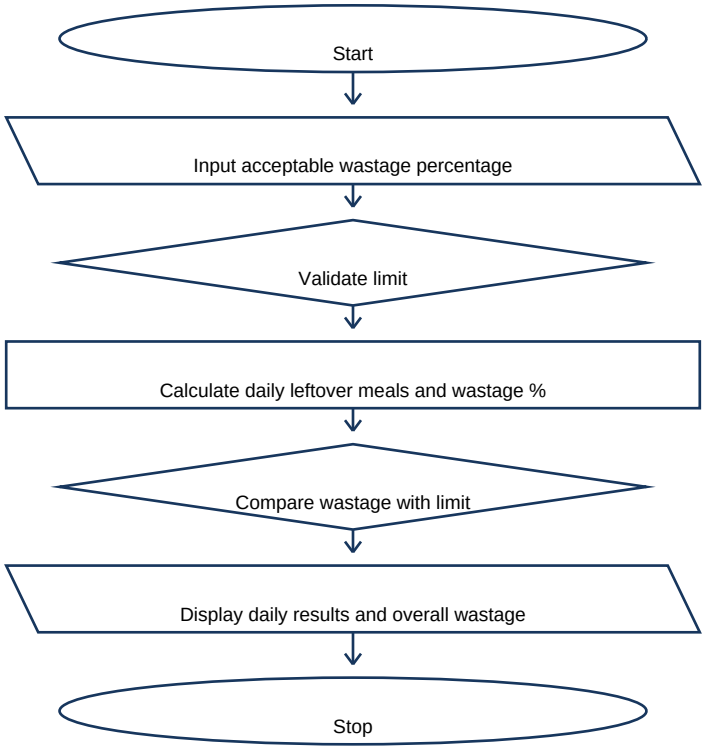
limit = float(input("Enter acceptable wastage percentage: "))

if not 0 <= limit <= 100:
    print("Invalid wastage limit")
else:
    total_waste = 0
    total_prepared = 0

    for i in range(len(days)):
        waste = prepared[i] - served[i]
        percentage = waste / prepared[i] * 100

        total_waste += waste
        total_prepared += prepared[i]
```

FLOWCHART



OUTPUT

```
Enter acceptable wastage percentage: 32

Day: Monday
Leftover meals: 10
Wastage percentage: 10.0 %
Wastage within limit

Day: Tuesday
Leftover meals: 20
Wastage percentage: 16.67 %
Wastage within limit

Day: Wednesday
Leftover meals: 5
Wastage percentage: 4.55 %
Wastage within limit

Day: Thursday
Leftover meals: 20
Wastage percentage: 22.22 %
Wastage within limit

Day: Friday
Leftover meals: 15
Wastage percentage: 11.54 %
Wastage within limit

Total meals wasted: 70
Overall wastage: 12.73 %
```